

Osman Akar

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EDUCATION

- PhD in Applied Mathematics at University of California, Los Angeles** 2019 - Present
Research Interests: Physics/PDE based computer graphics for animations and simulations, Graph based machine learning algorithms, Image processing
- Masters in Mathematics at University of California, Los Angeles** 2016 - 2019
Departmental Scholars Program
- BS in Mathematics of Computation at University of California, Los Angeles** 2015 - 2019
Math Undergraduate Merit Scholar, Graduated Cum Laude

RESEARCH EXPERIENCE

- Research Intern at Research in Industrial Projects for Students (RIPS)** Jun-Aug 2018
Institute for Pure and Applied Mathematics (IPAM) Los Angeles, CA
- Worked on a research team that build automated video augmentation pipeline that identifies crowd regions in sport videos and overlays an advertisement onto these regions, so that the advertisement looks aesthetic and natural for human perspective. [This video link](#) is an example of the final product. More explicitly, the automated video augmentation pipeline is able to (1) identify non-homogeneous textures (i.e., crowd regions in sports stadium imagery), (2) overlay a rectangular image asset on said region, and (3) constrain the asset to be placed in a perspective correct way for an overall enhanced user experience. Here is the resulting paper [arXiv:1907.09394](#).
 - Designed and implemented an algorithm to visualize RGB-Depth image frame in 3D. Implemented *Random Sample Consensus Algorithm* in this 3D-reconstruction to detect planes to overlay advertisement onto the image in 2D.
 - Developed a novel metric on video frames for the evaluation of the *PSPNet* crowd segmentation algorithm.
- Undergraduate Researcher at Computational and Applied Mathematics REU** Jun-Aug 2017
California Research Training Program in Computational and Applied Mathematics Los Angeles, CA
- Worked on a research team that applied image processing techniques to extract non-content-based features from LAPD body-worn-camera video recordings and used machine learning to classify the recordings into categories.
 - Modified the old machine learning model to accept the new dataset of LAPD videos. Experimented with tuning hyperparameters and reduced the error by 50%. Here is the resulting paper [arXiv:1904.09062](#)
 - Created color-based feature and its metric system for video frames and implemented in the ML Algorithm.

TEACHING EXPERIENCE

- Teaching Assistantship at UCLA** Sep 2019-Current
- TAed the following courses: [Introduction the Probability](#) and Graduate Level [Applied Differential Equations](#)
 - Led class of undergraduate students in the weekly discussion sections. Hold office hours and helped the instructor with grading and proctoring the exams.
- Math Olympiads Lecturer & Curriculum Designer at Star League, Inc** Sep 2017 - April 2020
- Lectured math olympiad courses in AMC 8/10/12 and AIME levels throughout 12-week semester-long sessions and 3-week summer camps. Presented problems from earlier years' math olympiads competitions.
 - Organized curriculum and problem set for the 3-week AIME preparation summer program.

LIST OF HONORS AND AWARDS

- Putnam Performance Prize by UCLA Mathematics Department, 2019
- Honorable Mention at [William Lowell Putnam Mathematical Competition](#), 2018. Ranked 77 among 4623 students.
- Richard F. Arens Putnam Scholars Undergraduate Award by UCLA Mathematics Department, 2016
- Sole recipient of [UCLA Math Undergraduate Merit Scholarship](#) in class of '19
- [Gold Medal](#) at International Mathematical Olympiad, 2014, South Africa
- Special Prize at European Union Contest for Young Scientist, 2014, Poland
- Gold Medal with perfect score at Balkan Mathematical Olympiad, 2014, Bulgaria
- Silver Medal at International Mathematical Olympiad, 2013, Colombia

PROGRAMMING & TECHNICAL

Languages: PYTHON, MATLAB, C++, L^AT_EX **Familiar:** C, JAVASCRIPT, ASSEMBLY